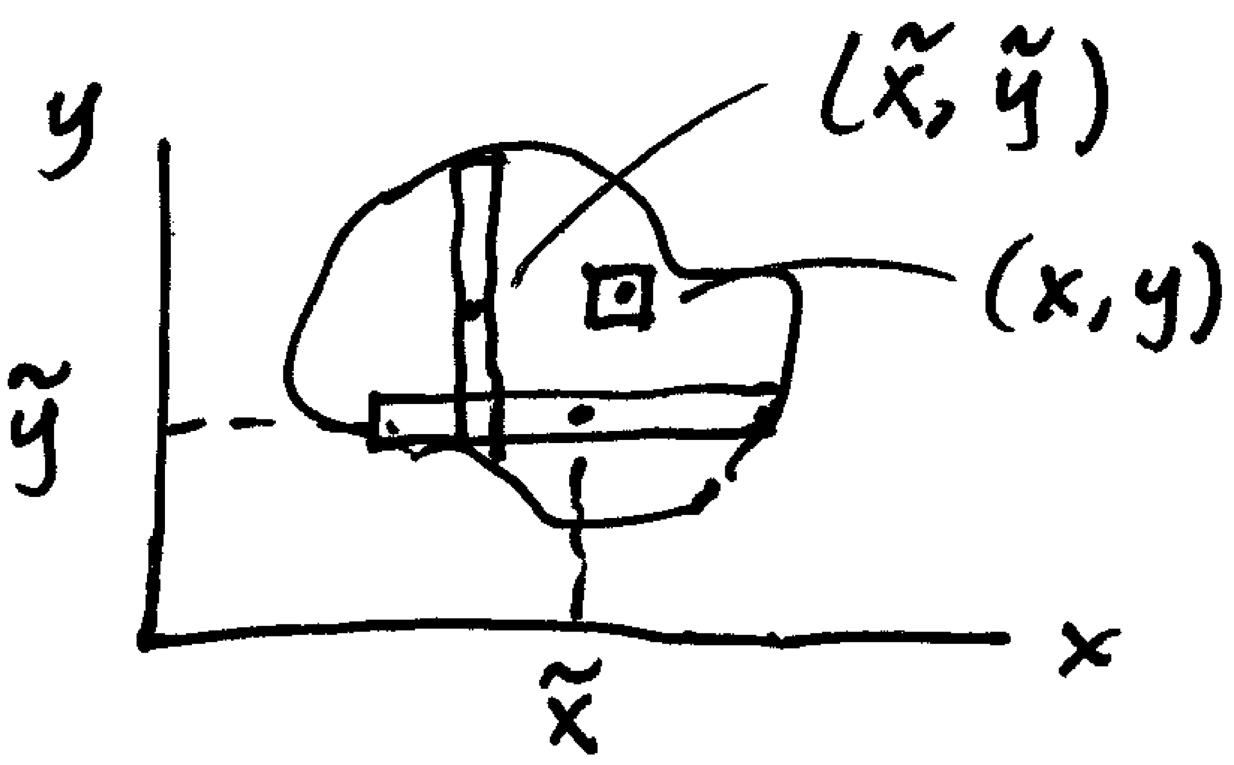
Hibbeler Crib Sheet

Area Centroid

$$\boxed{\bar{x} = \frac{\int_A \tilde{y}\, dA}{\int_A dA}}$$

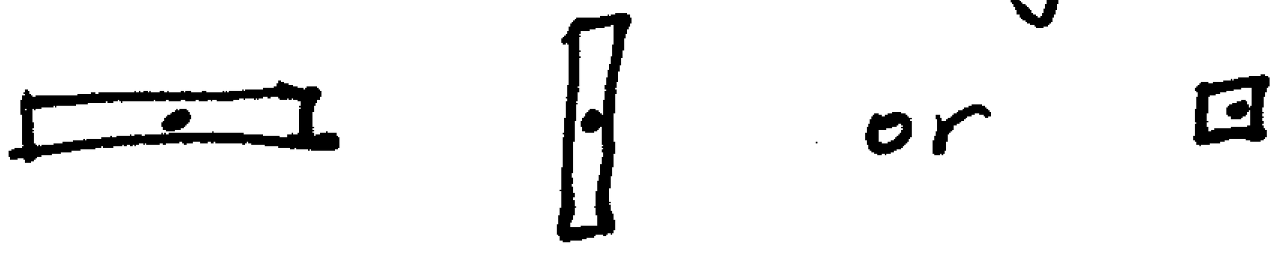


where, $\tilde{x}$ = dist to centroid of dA

$\tilde{y}$ = " " " " "

$$\boxed{\bar{y} = \frac{\int_A \tilde{x}\, dA}{\int_A dA}}$$

You can use any dA



as long as define $(\tilde{x}, \tilde{y})$ correctly.

BUT note on the Hibbeler crib sheet:

For Area MOI (Moment of Inertia)...

He defines these as:

$$\boxed{I_x = \int_A y^2\, dA}$$

$$\boxed{I_y = \int_A x^2\, dA}$$

He does not use $\tilde{x}$ or $\tilde{y}$!!

Why?

Short answer: You cannot use these definitions for I_x and I_y with any dA like we did for centroids!

Be careful!

Easiest dA's to use for Area MOI's:

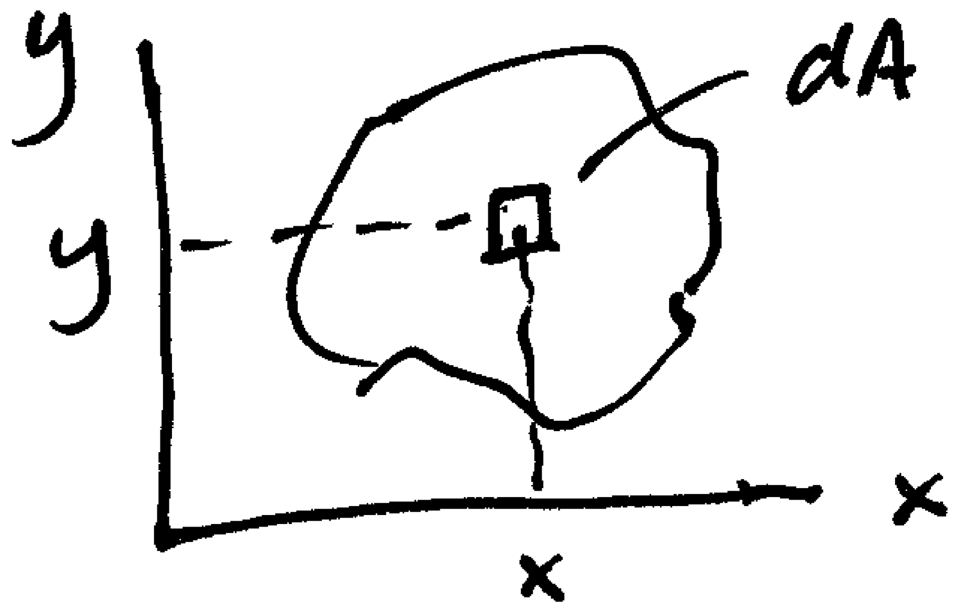
1. use $dA = dx\,dy$ or $dA = dy\,dx$ and double integral.

This is the easiest and best, in my opinion.

Then,

$$\boxed{I_x = \int_A y^2\, dA}$$

$$\boxed{I_y = \int_A x^2\, dA}$$



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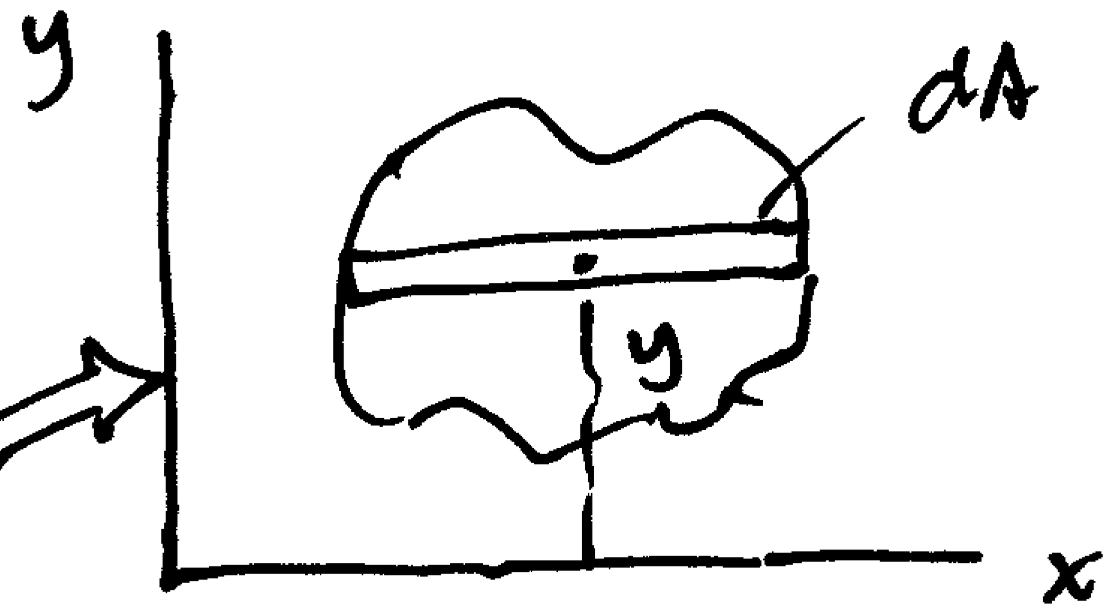
2. Another good choice is to use dA's parallel to the axes:

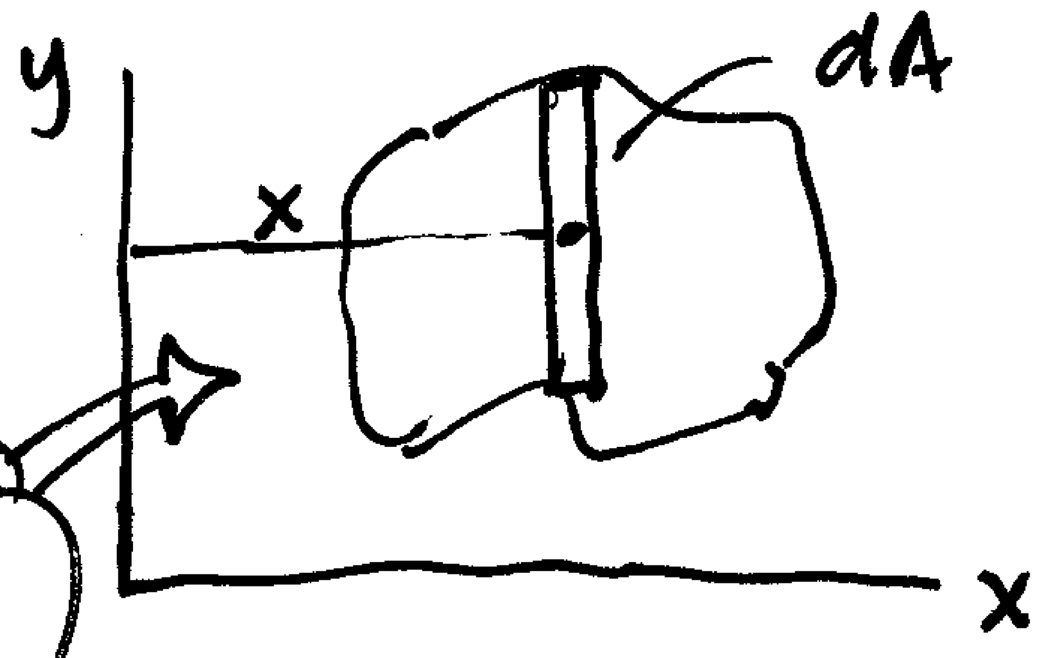
thus,



Entire element is dist y from x axis

$$\boxed{I_x = \int_A y^2\, dA}$$

and

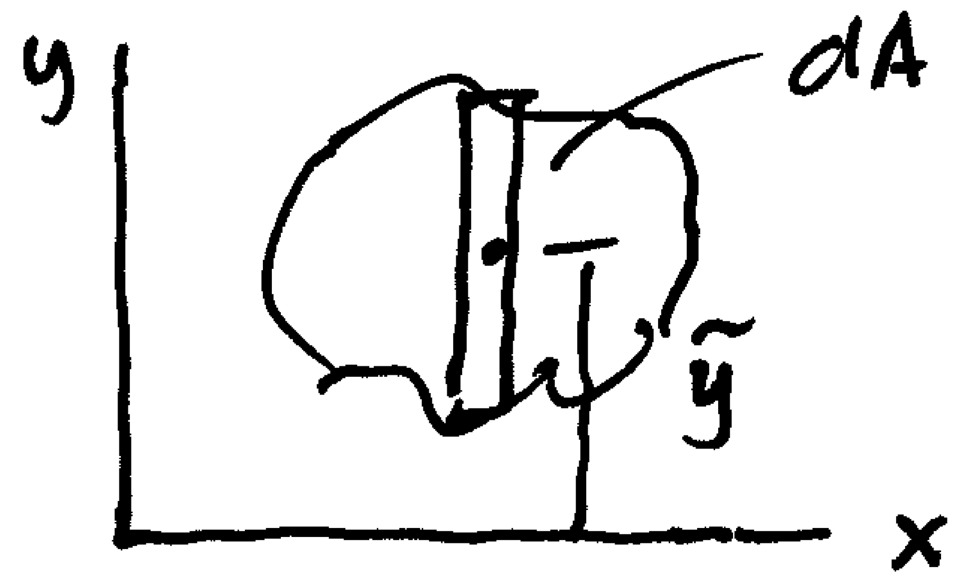


Entire element is dist x from y axis

$$\boxed{I_y = \int_A x^2\, dA}$$

③ But can we use dA's $\perp$ to the axis of interest?

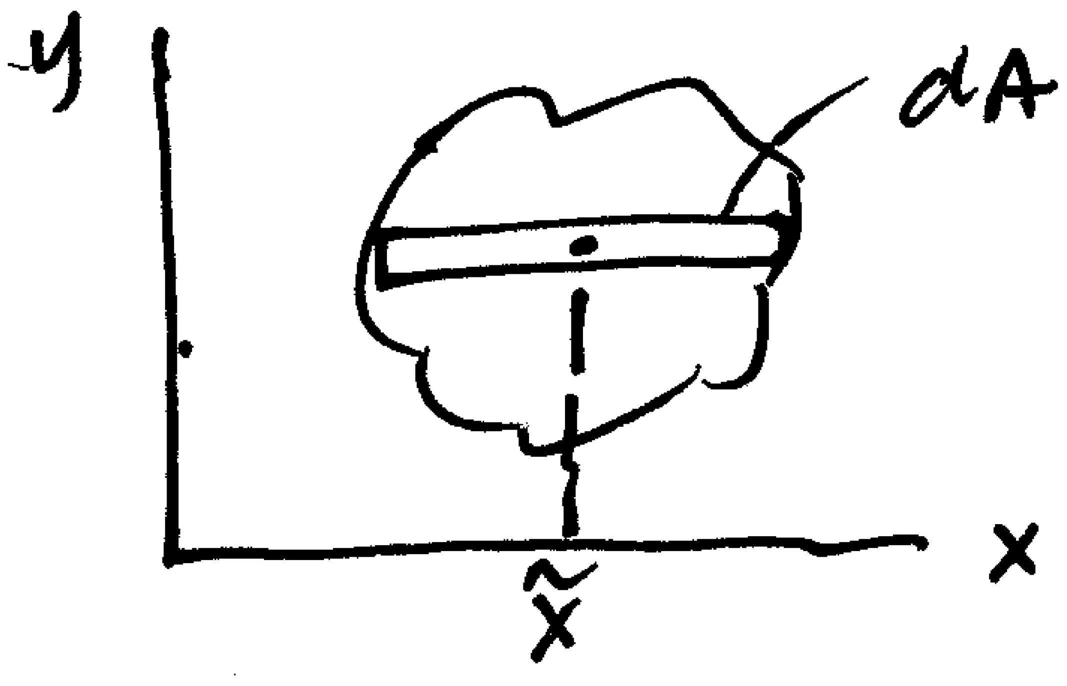
e.g.



$$I_x = \int_A \tilde{y}^2 dA \ ?$$

NO! This is not valid!

or



$$I_y = \int_A \tilde{x}^2 dA \ ?$$

NO! Not valid!

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④ Why do these dA's NOT work, but the others do?

Short answer: when we write

$$\boxed{I_x = \int_A y^2 dA} \quad \text{or} \quad \boxed{I_y = \int_A x^2 dA}$$

* with the proper dA's !!

we are summing the MOI's of the dA's about the respective axes.

In other words, we are really writing...

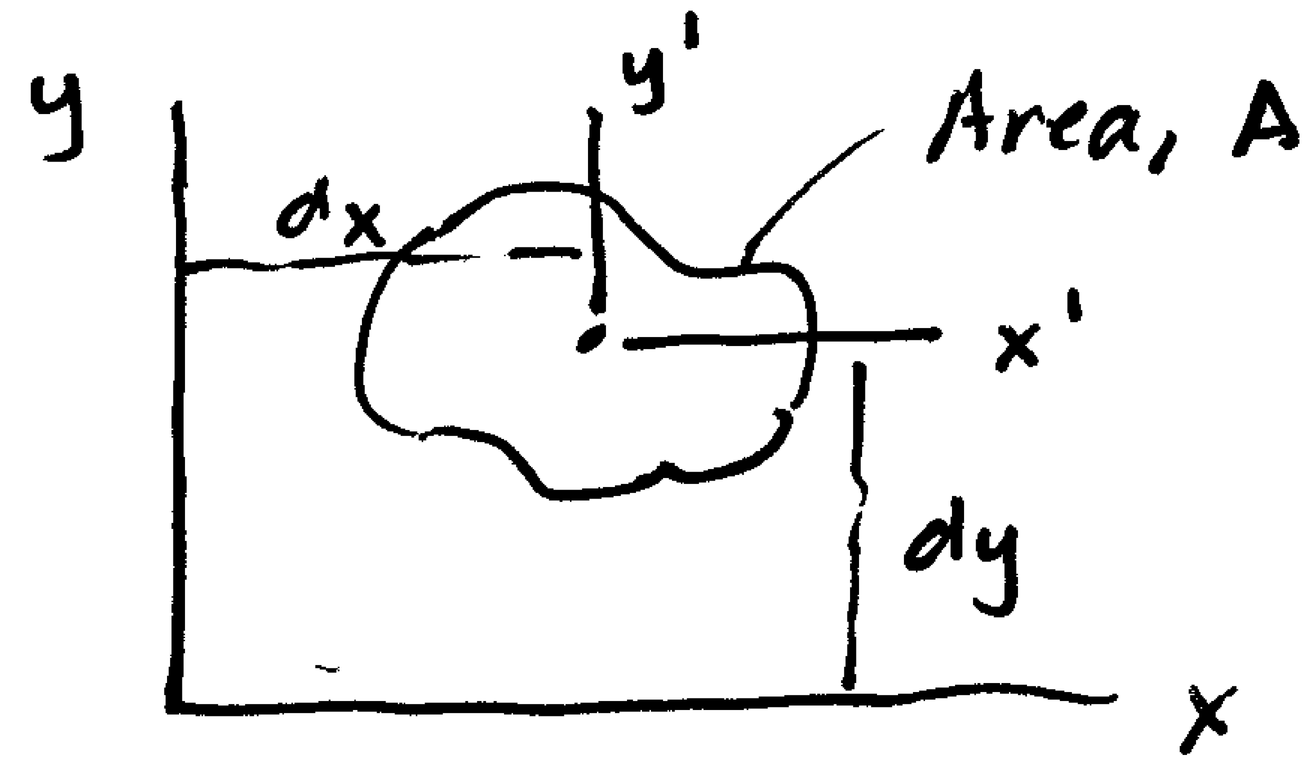
$$I_x = \int dI_x,$$ where dI_x = MOI of the dA about the x axis

and

$$I_y = \int_{,} dI_y,$$ where dI_y = MOI of the dA about the y axis

5 Which leads us to the <u>Parallel Axis Theorem</u> (PAT)

(will derive the PAT soon...)



What is the PAT?

$$\boxed{I_{Axis} = \bar{I} + Ad^2}$$

It's used to "transfer" a MOI about an area's centroidal axis ($\bar{I}$) to another axis parallel to it.

x' and y' are centroidal axes..

Thus, for our area, A, above,

if we know $\bar{I}_{x'}$, A, and dy,

then $\boxed{I_x = \bar{I}_{x'} + A \cdot dy^2}$

and, if we know $\bar{I}_{y'}$, A, and dx,

then $\boxed{I_y = \bar{I}_{y'} + A \cdot dx^2}$

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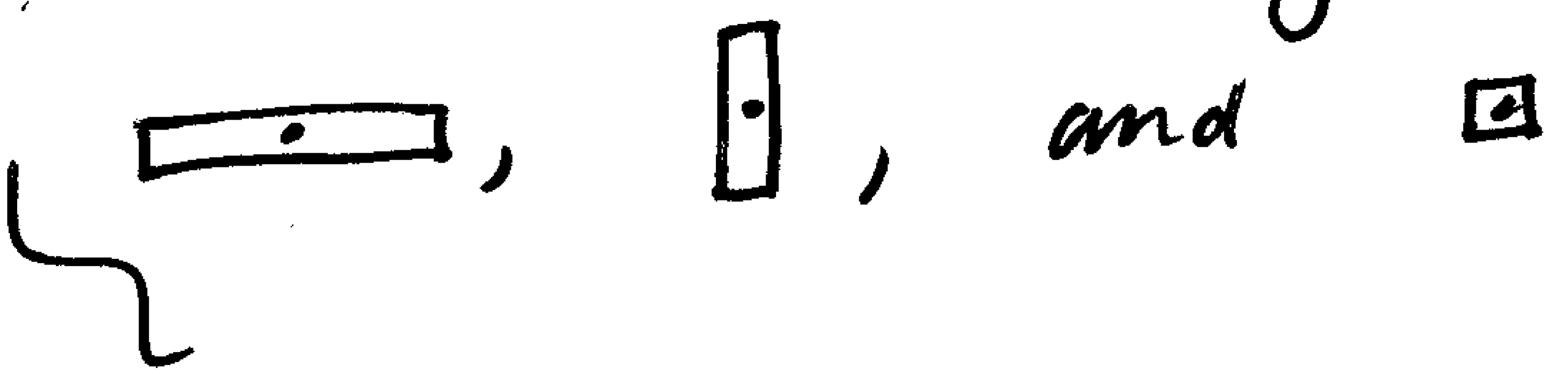
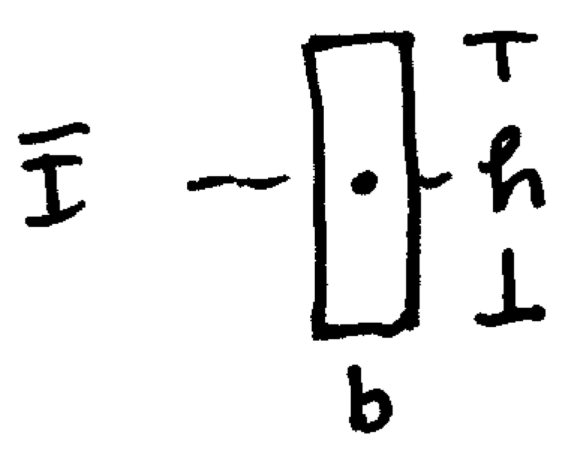
6 So how do these apply to



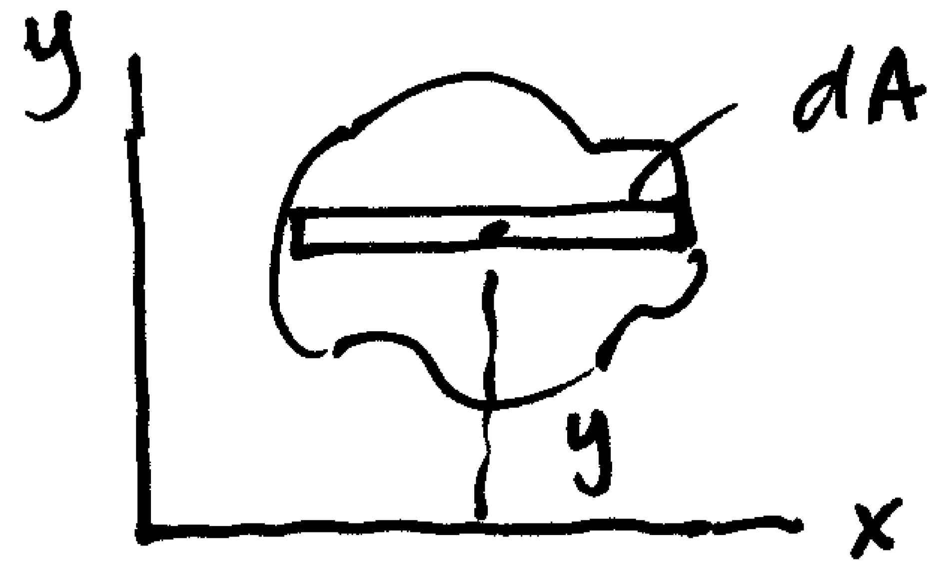
and dA's?

All of these dA's are rectangles...



for a general rectangle <u>about its centroidal axis</u>, its $\boxed{\bar{I} = \frac{1}{12} b h^3}$

7 So how does this PAT apply to various dA's?



If our most general formula for MOI is

$$I_x = \int dI_x$$

we can use the PAT in differential form to find I_x of the dA!

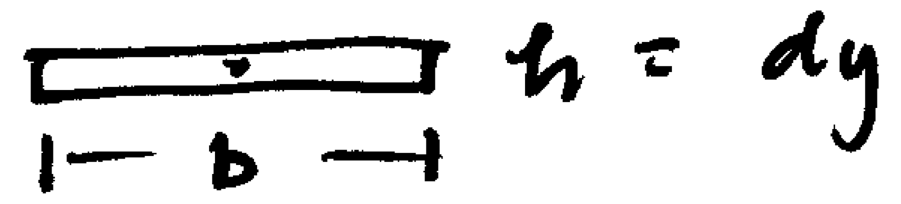
PAT in differential form:

$$\boxed{dI_x = d\bar{I}_{x'} + dA \cdot \tilde{y}^2}$$

and

$$\boxed{dI_y = d\bar{I}_{y'} + dA \cdot \tilde{x}^2}$$

For the case above,



thus, $d\bar{I}_{x'} = \frac{1}{12} b (dy)^2 = 0$!

thus, $dI_x = y^2 \, dA$

finally,

$$\boxed{I_x = \int dI_x = \int y^2 \, dA}$$

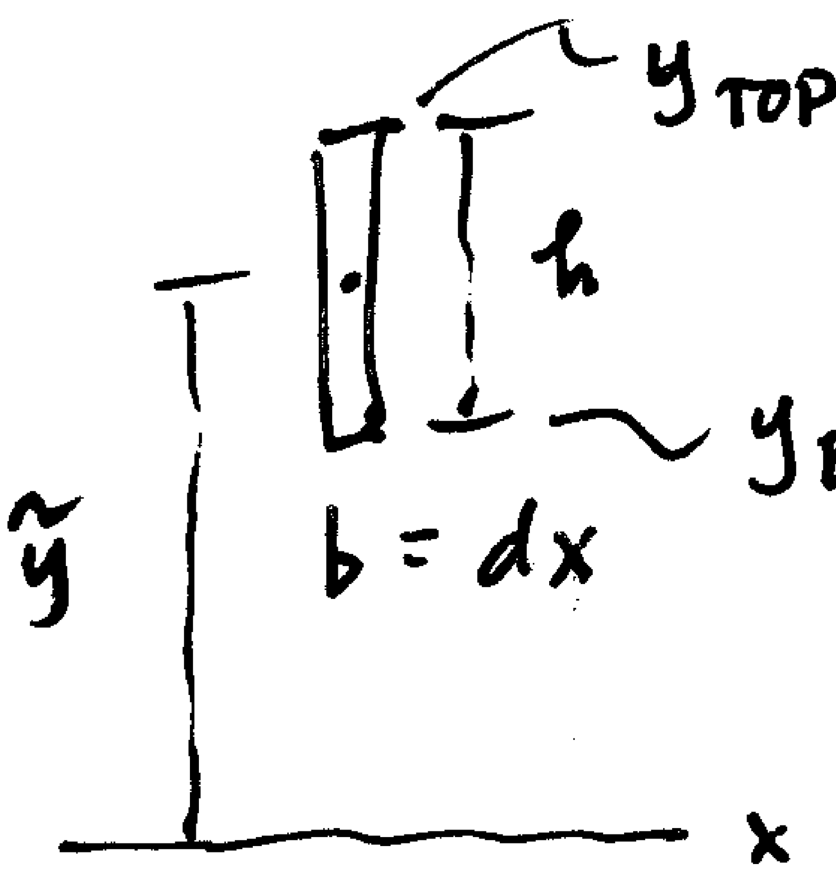
for an element parallel to the x axis!

8 What about an element $\perp$ to the x axis?

We still write $\quad I_x = \int dI_x$

__But__ what is dI_x?

Must use the PAT in differential form:



$$d\bar{I}_{x'} = \frac{1}{12} b h^2$$

$$= \frac{1}{12}(dy)(y_{TOP} - y_{BOT})^2 \quad !$$

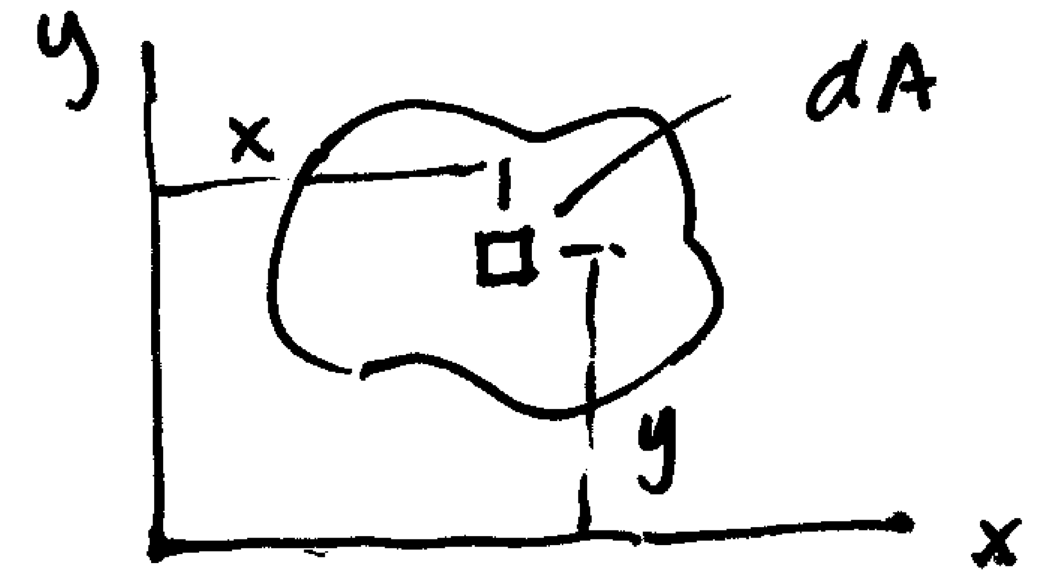
__and__ $$\tilde{y} = \left(\frac{y_{BOT} + y_{TOP}}{2}\right) \quad !$$

Thus,
$$I_x = \int dI_x = \int \left(d\bar{I}_{x'} + dA \cdot \tilde{y}^2 \right)$$

$$I_x = \int \left[\frac{1}{12}(y_{TOP} - y_{BOT})^2 dx + \left[\frac{(y_{BOT} + y_{TOP})}{2}\right]^2 \cdot \underbrace{(y_{TOP} - y_{BOT})\, dx}_{\text{this is } dA} \right]$$

9 What about an element
$dA = dx\,dy$ or $dA = dy\,dx$?

Use the PAT in differential form:



$$dI_x = d\bar{I}_{x'} + y^2\,dA$$

dx dy $$d\bar{I}_{x'} = \frac{1}{12}bh^2$$
$$= \frac{1}{12}(dx)(dy)^2 \Rightarrow 0$$

thus, $I_x = \int dI_x$

$$\boxed{I_x = \int y^2\,dA}$$ as expected ...

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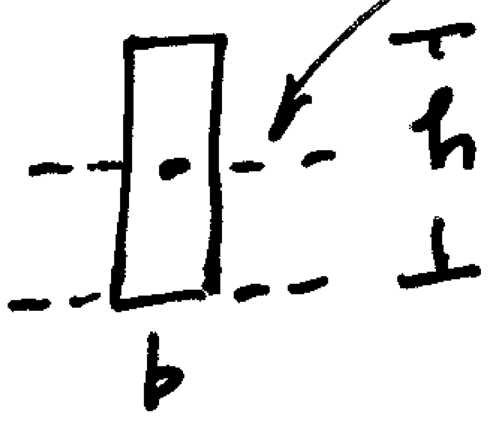
10 One other case: Can use a $\perp$ element if it touches the axis.

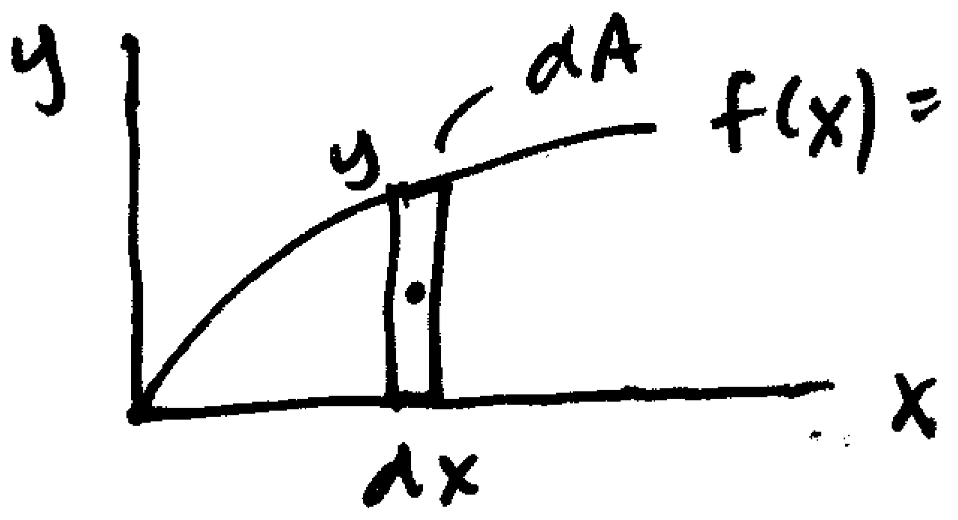
How? For a rectangle,



$$\boxed{\bar{I} = \frac{1}{12}bh^3}$$

$$\boxed{I_{base} = \frac{1}{3}bh^3}$$

So, if have an area which touches the x-axis



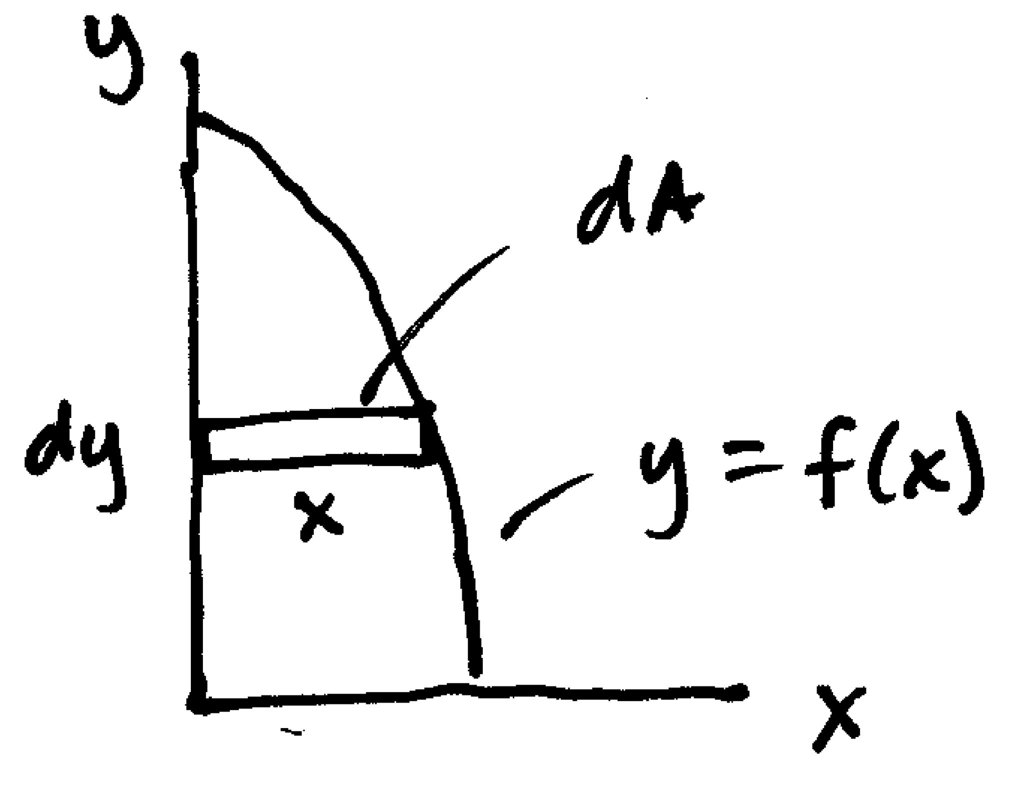
$$I_x = \int dI_x$$

BUT, $$dI_x = \frac{1}{3}bh^3$$
$$= \frac{1}{3}(dx)\,y^3 \;!$$

and $$\boxed{I_x = \int \frac{1}{3}y^3\,dx}$$

$y = f(x)$

11 Similarly, for I_y, if area touches the y axis...



$$I_y = \int dI_y$$

But $dI_y = \frac{1}{3} b h^3$

$$= \frac{1}{3} (dy)\, x^3$$

convert to y function

Finally, $\boxed{I_y = \int \frac{1}{3} x^3 \, dy}$

convert to y function